

SuSiE-mix: simulation sections

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Results

Simulation benchmark using observed genotype structure

We compared additive SuSiE with unweighted SuSiE-mix using simulated quantitative phenotypes and observed GTEx genotypes. The benchmark used 500 donors per dataset, one to five distinct causal SNPs, and target proportions of variance explained (PVE) of 10%, 20%, 30%, and 40%. We enumerated all 55 allocations of these causal SNPs to additive, dominant, and recessive effects, giving 220 allocation-PVE conditions with 400 scheduled replications each. The completed archive contained 87,988 usable datasets and 175,976 converged fits; 12 scheduled datasets had zero genetic variance and were excluded without replacement. Both methods were fitted to the same phenotype with an upper bound of ten single-effect components. SNP-localization analyses combined the retained encodings of each SNP in the mixed-model output. We additionally evaluated exact identification of the generating SNP-coding pair using the global lead PIP (Methods).

The seven architecture classes comprise the three single-coding models, the three possible pairs of codings, and all three codings together. Within each class, results at a given causal count pool every compatible allocation. Coverage figures retain the causal count on the horizontal axis; pooled PIP curves combine all causal counts within each architecture and PVE. The same 400 replication seeds were reused across conditions, so the condition-specific results are dependent. The figures report pooled point estimates; uncertainty intervals for the coding PIP-mass shares resample these shared replication blocks (Methods).

Non-additive effects reduce coverage and causal-SNP recovery

Allowing alternative genotype encodings improved credible-set coverage under non-additive generating models, with the largest gains under recessive effects (Figs. 1 and 2). For example, with three recessive causal SNPs and PVE of 20%, additive SuSiE achieved 83.0% empirical coverage, whereas SuSiE-mix achieved 96.5%. The fraction of causal SNPs recovered in the union of reported credible sets increased from 21.2% to 79.2% (Fig. S1). At the same causal count and PVE under dominant effects, coverage increased from 92.2% to 95.4%, and credible-set recovery increased from 69.5% to 78.8%.

Gains also occurred when different causal SNPs followed different encodings. With five causal SNPs spanning all three encodings and PVE of 40%, coverage increased from 87.3% to 93.7%, and credible-set recovery increased from 61.1% to 85.9% (Figs. 2 and S2). These results show that the effect of additive misspecification extends beyond loss of sensitivity: reported credible sets can also omit every generating causal SNP more often than the nominal 5% rate.

PIP-based detection at comparable empirical false discovery proportions

PIP-based comparisons supported improved discrimination under non-additive effects (Figs. 3 and 4). Pooling all causal counts at PVE of 40%, recessive-model detection power near 10% empirical FDR increased from 6.1% with additive SuSiE to 31.2% with SuSiE-mix. Corresponding power increased from 22.0% to 30.5% under dominant effects and from 11.5% to 24.8% for architectures containing all three effect types. Under additive-only effects, the corresponding powers were similar, 26.9% and 26.6%. These descriptive operating points use the last sampled PIP threshold before the first crossing of 10% empirical FDR as the threshold decreases (Methods).

The ROC curves likewise showed the clearest separation under recessive effects, while the additive-only curves were close (Figs. S5 and S6). ROC curves stratified by causal count are provided in Figs. S7-S15. The absolute power at low empirical FDR remained limited, even where ROC power was high: the false positive rate divides false selections by all noncausal SNPs, whereas empirical FDR divides them by selected SNPs. The power-FDR comparison therefore addresses a different and more selective operating regime. Here empirical FDR is the pooled false discovery proportion across simulated datasets; these curves do not establish formal FDR control or PIP calibration.

Additive-case costs and remaining undercoverage

The expanded model did not improve every performance measure. Under additive-only effects with three causal SNPs and PVE of 20%, credible-set recovery decreased from 77.6% to 74.7%, despite similar coverage (93.1% versus 93.5%). Across all additive-only conditions, pooled credible-set recovery was 65.7% for SuSiE and 63.3% for SuSiE-mix. Furthermore, SuSiE-mix did not uniformly attain nominal coverage. With five causal SNPs spanning all three encodings and PVE of 10%, coverage improved from 81.9% to 87.6% but remained below 95%, and credible-set recovery was only 7.5% and 8.7%, respectively.

Purity was also architecture dependent. For the three-recessive-SNP, 20%-PVE example, mean minimum absolute within-set correlation increased from 0.820 to 0.911; under the corresponding additive-only scenario it decreased from 0.893 to 0.861 (Figs. S3 and S4). Mixed-model purity is computed on fitted coding-specific predictors, so it is not a common additive-LD measure of resolution. These results support a benefit from modeling non-additivity while showing that broader predictor models retain power and calibration trade-offs.

High-PIP leads identify generating SNP-coding pairs

To assess whether non-additive labels identify the generating coding, we selected the single largest coding-specific PIP across each simulated locus and evaluated the exact SNP-coding pair. Among unambiguous leads with $\text{PIP} \geq 0.9$, 6,521 of 6,805 dominant leads (95.8%) and 11,361 of 11,570 recessive leads (98.2%) matched a generating pair; the corresponding rate for additive leads was 5,458 of 6,063 (90.0%; Table 1). Errors at a causal SNP with the wrong coding accounted for 92 dominant leads, six recessive leads, and 100 additive leads; the remaining errors selected noncausal SNPs.

Table 1. Exact discovery among high-PIP global leads.

Called coding	Leads	Exact matches	Mean PIP (%)	Exact (%)
Additive	6,063	5,458	97.7	90.0
Recessive	11,570	11,361	98.9	98.2
Dominant	6,805	6,521	97.8	95.8

One positive, unambiguous global lead was selected per dataset; only leads with coding-specific PIP ≥ 0.9 are included. An exact discovery matches both a generating causal SNP and its generating coding. Counts pool all causal allocations, PVE values, and causal counts with one observation per selected dataset. Mean PIP and exact-match frequency are percentages. These rates measure precision of the selected leads, not recovery of all causal effects.

Exact-match frequency generally increased with lead PIP (Fig. 5). However, numerical PIPs were not uniformly calibrated on this simulation grid. Mean PIPs among the high-PIP additive, dominant, and recessive leads were 97.7%, 97.8%, and 98.9%, respectively, compared with exact-match frequencies of 90.0%, 95.8%, and 98.2%. Performance also depended on signal strength and causal count: at PVE of 10% with five causal effects, only 12 of 26 high-PIP dominant leads were exact matches; all 14 errors selected noncausal SNPs. Condition-specific counts are supplied with the simulation tables. Thus, the pooled accuracy does not imply uniformly reliable inference in weak, multicausal loci.

Additive controls distinguish confident calls from diffuse coding support

The 7,996 additive-only datasets provided a control for false non-additive lead calls. Only 27 datasets (0.34%) produced an unambiguous dominant global lead with PIP ≥ 0.9 , and three (0.038%) produced a recessive global lead at this threshold. Without a PIP threshold, dominant coding accounted for 11.9-27.9% of unambiguous leads across causal counts at PVE of 10%, but 0.6-5.8% at PVE of 40% (Fig. S16). These per-dataset control rates support the specificity of confident non-additive lead calls within the evaluated design, while showing that the highest-ranked coding alone is less informative when posterior support is weak.

Confident lead calls and pooled posterior mass nevertheless gave different summaries. Even under wholly additive generating models, the sum of dominant PIPs divided by the sum of PIPs over all codings ranged from 18.0% to 35.0% across PVE and causal-count conditions; the corresponding recessive shares ranged from 6.6% to 24.9% (Fig. S17). These nonzero shares show that pooled coding PIP mass cannot directly be interpreted as the fraction of generating effects following each coding. The lead results provide support for investigating high-confidence non-additive candidates in the main analysis, while the mass-allocation controls limit conclusions about their genome-wide prevalence.

Methods

Simulation design and genotype sampling

The completed benchmark was generated by `sim_workhorse.R` and the R jobs written by `write_jobs.R`. We fixed the sample size at $n=500$ and the fitted upper bound at $L_{\text{fit}}=10$. For every true causal count $K=1,\dots,5$, we enumerated all nonnegative integer triples (K_A, K_R, K_D) summing to K , where zero excludes that generating coding. There are 55 such allocations. Each was crossed with PVE h in $0.10, 0.20, 0.30, 0.40$, yielding 220 conditions with 400 scheduled replications per condition. Single-coding architectures permit $K=1,\dots,5$, two-coding architectures $K=2,\dots,5$, and the three-coding architecture $K=3,\dots,5$. The true count K is denoted by "L" in the PIP figure titles; it is distinct from the fixed fitted bound of ten components.

For each replication, one pre-extracted GTEx PLINK genotype file was sampled uniformly from the available .raw files. Thus, the sampling frame was the available genotype-file collection rather than a new draw from a genome-wide gene list. SNPs with any missing genotype or no variation were removed. Across all donors in that file, genotypes were oriented to the minor allele, SNPs were required to have minor-allele frequency strictly greater than 0.05, and SNPs with a Pearson Hardy-Weinberg chi-square P value below 10^{-8} were excluded. We then sampled 500 distinct donors without replacement, retaining the full-donor allele orientation and local genotype correlations. No additional MAF/HWE filtering or LD pruning was applied to this donor subset.

For minor-allele dosage G_{ij} in $0,1,2$, the candidate predictors were G_{ij} (additive), $1G_{ij} \geq 1$ (dominant), and $1G_{ij} = 2$ (recessive). Each coding was filtered separately to require positive sample standard deviation and a column sum of at least five. This requires five allele copies for additive predictors, five carriers for dominant predictors, or five minor-allele homozygotes for recessive predictors; it does not require five donors in every genotype class. Consequently, the retained coding blocks can have different numbers of columns.

The causal-SNP pool was the intersection of the retained additive SNPs and the retained SNPs of every non-additive coding requested by the allocation. We sampled K distinct biological SNPs uniformly without replacement from this common pool and assigned the first K_A additive, the next K_R recessive, and the final K_D dominant effects. Each causal SNP had exactly one generating coding, and there were no interactions. In conditions containing recessive effects, this common eligibility rule also restricts the additive and dominant causal SNPs to those retaining a recessive predictor. An ineligible draw or a zero-variance genetic signal produced a saved error without drawing a replacement locus. No tissue-specific expression phenotype, covariate adjustment, or inverse-normal transformation was used in the simulation.

Effect sizes, LD and phenotype generation

Let W_{ik} be the generating encoding of causal SNP k in donor i , and let W_k and s_k be its sample mean and standard deviation. We standardized each causal predictor and independently sampled signs a_k in $-1,1$ with equal probabilities:

$$z_{ik} = (w_{ik} - \text{mean}(W_k)) / s_k; \quad g_i^{(\theta)} = \sum_k a_k z_{ik}.$$

For target PVE h , we applied one common rescaling to all effects:

```
b = sqrt[h / sample_var(g^(0))]; g_i = b g_i^(0); y_i = g_i + error_i,
with independent error_i ~ Normal(0, 1 - h).
```

Sample variance uses denominator $n-1$. This gives sample $\text{Var}(g)=h$ while accounting for covariance among causal predictors. Each individual contribution $b a_k z_k$ has the same variance b^2 ; in LD this is not an allocation of h/K to each SNP, because the variance of the sum also includes covariance terms. The target PVE is defined as sample genetic variance divided by sample genetic variance plus the generating noise variance. Realized sample $\text{Var}(g)/\text{sample Var}(y)$ can differ from h because the noise is an independent Gaussian draw. Errors were neither orthogonalized against the genotypes nor rescaled after drawing.

Model fitting and SNP-level evaluation

We fitted `susieR::susie` to the retained additive matrix and to the concatenated additive, recessive, and dominant matrices. Both fits used the same outcome, `L_fit=10`, `standardize = TRUE`, `estimate_prior_method = "optim"`, `coverage = 0.95`, `min_abs_corr = 0.5`, and `max_iter = 1000`. The true causal count, generating codings, and residual variance were not supplied. Other controls used the installed package defaults. The benchmark compares ordinary additive and unweighted mixed fits, not the weighted or EM-reweighted coding models. Uniform prior inclusion weights are over retained predictor columns; in the mixed fit, total prior weight per biological SNP therefore depends on the number of encodings surviving filtering. The standardization, prior-variance optimizer, and purity threshold differ from the GTEx analysis settings described above.

For SuSiE-mix, coding-specific posterior assignment probabilities were summed within each single-effect component before calculating a PIP for biological SNP j :

```
PIP_j = 1 - product_l [1 - sum_(c in C_j) alpha_(l,j,c)], over active
components l.
```

Here C_j is the set of retained encodings of SNP j . The fitted prior variances were retained when calling `susie_get_pip`, so its usual exclusion of inactive components was preserved. Final coding-specific PIPs were not summed. Mixed credible sets were mapped to unique biological SNP indices within each set, without reconstructing new credible sets from SNP-level PIPs. A reported set was considered covered if it contained at least one generating causal SNP, irrespective of the member's encoding. This evaluates SNP localization, not correct identification of a generating coding or a one-to-one correspondence between components and causal SNPs.

Empirical coverage was the total number of covered reported sets divided by the total number of reported sets. Credible-set recovery was the number of generating causal SNPs appearing in the union of reported sets divided by the total number of generating causal SNPs; each SNP was counted once within a dataset. Runs with no reported set contributed zero recovery and no sets to the coverage or purity denominator. Purity was the mean of the saved minimum absolute within-set correlations, using the fitted predictor encodings for each method. Singleton purity was one. We did not recompute mixed-set purity on additive genotypes. Coverage and purity therefore weight reported sets, whereas recovery weights causal SNPs.

PIP curves, aggregation and computational record

At each common PIP threshold t , a SNP was selected when its PIP was at least t . We summed true positive (TP) and false positive (FP) counts across the included datasets and calculated power as TP divided by all causal SNPs, false positive rate as FP divided by all noncausal SNPs, and pooled empirical FDR as FP divided by TP plus FP. The last quantity is a pooled false discovery proportion, not the unweighted mean of per-dataset false discovery proportions. The no-discovery endpoint was defined as zero power and zero empirical FDR.

Counts were evaluated on a common dense grid with steps of 0.001 between 0.01 and 0.99, logarithmically spaced tails down to 10^{-12} and up to $1 \cdot 10^{-12}$, and endpoints at zero, one, and infinity. Tied PIPs entered together. Curves join threshold-specific points in decreasing threshold order; empirical-FDR curves were not sorted by FDR or replaced by an optimized envelope. The numerical comparisons near 10% empirical FDR use the grid point immediately preceding the first exceedance of 0.10 in this order. Thresholds and achieved rates are supplied with the summary tables. These truth-based operating points are descriptive evaluations, not a proposed threshold-selection rule for data with unknown causal variants.

For plots at fixed K , all compatible allocations within an architecture were pooled. For "All L pooled" plots, TP and FP counts were also summed across all available K before rates were recalculated. This is equivalent to stacking the SNP observations; it does not average per- K rates or give each K equal weight. Larger loci contribute more noncausal SNPs, and higher causal counts contribute more causal SNPs. In mixed architectures, causal counts admitting more allocations also contribute more datasets. Available-case weights reflect the small number of failed draws.

Replication r used seed $1,000,000+r$, for $r=1,\dots,400$, with the same seeds reused across allocations and PVE values. The archive comprised 221 checkpoints: 220 production-condition files and an older overlapping checkpoint. Nine duplicate condition-seed records were removed. Of 88,000 scheduled condition-replication records, 12 had zero genetic variance and 87,988 were usable; no included fit was flagged as nonconverged. Errors were omitted from both methods rather than counted as no-discovery outcomes. Saved objects contain settings, seeds, the genotype-file path, causal SNPs and codings, standardized effects, convergence status, credible sets, and PIPs; they do not retain the simulated outcome, sampled donor indices, or full fitted models.

Coverage, recovery, ROC, power-FDR, and lead-PIP figures display pooled point estimates without Monte Carlo uncertainty intervals. The coding PIP-mass figure additionally reports seed-block bootstrap intervals, as described below. Shared seeds define replication blocks across conditions and must be retained together in uncertainty analyses. The current figure set does not include a credible-set-size comparison. The analysis scripts, plotting code, checkpoint audit, configuration counts, and numerical summaries document the reported results. The cluster's R and package versions and the immutable genotype-file inventory were not recorded in these compact objects and remain to be added to the final reproducibility record.

Exact discovery and reliability of the global lead PIP

We used the saved coding-specific PIPs from the unweighted SuSiE-mix fit to select the largest PIP across all retained SNP-coding pairs in each dataset. Selection did not use generating truth, credible sets, or biological-SNP PIPs. A positive lead was unambiguous if no other predictor was within an absolute PIP difference of 10^{-12} of the maximum. Multiple such predictors were recorded as tied leads, including ties across distinct SNPs or between true effects; they were not resolved using column order or truth. Datasets with all PIPs zero were recorded separately. There were 15,547 tied datasets and 71 with zero support among the 87,988 usable datasets; only 20 tied datasets had maximum PIP ≥ 0.9 .

An exact discovery required the selected predictor to match both the SNP and coding of any generating effect. Errors were divided into a wrong coding at a causal SNP and selection of a noncausal SNP. For multiple causal effects, this evaluates the strongest reported hit; it does not measure recovery of every effect. For positive, unambiguous leads, we grouped maximum PIPs into ten bins, $[0,0.1), \dots, [0.9,1]$, and compared mean PIP with the fraction of exact matches by called coding and PVE. Figure 5 pools all causal allocations and counts within PVE, with one observation per selected dataset; tables also retain PVE and causal count separately. Empty bins have no estimated accuracy. Table 1 restricts these leads to PIP ≥ 0.9 and pools the full grid.

In additive-only datasets, we calculated the frequency of each called coding among positive, unambiguous leads. The dominant and recessive high-PIP control rates instead used all 7,996 usable additive-only datasets as the denominator, counting whether a dataset produced an unambiguous non-additive global lead with PIP ≥ 0.9 . These are rates of false non-additive lead calls per simulated dataset, not false discovery proportions among non-additive calls in real data. The checked, deduplicated checkpoint collection was the same as for the SNP-localization analysis. Unequal retained coding blocks were reconstructed from saved predictor-to-SNP maps and checked against the generating indices. PIP excursions outside $[0,1]$ no larger than 10^{-10} were clipped for numerical roundoff; this affected 24 datasets, including roundoff in biological-SNP PIPs. No dataset was excluded because of an invalid coding map or PIP.

Coding PIP-mass allocation under additive truth

For each additive-only PVE and causal-count condition, we pooled the raw coding-specific PIPs over all retained predictors and usable datasets. For coding c , the mass share was

$$p_{i,c} = \frac{\text{(sum of PIPs for coding } c \text{ over datasets and predictors)}}{\text{(sum of PIPs over all codings, datasets and predictors)}}$$

where P_r is the set of retained SNP-coding pairs in dataset r . No PIP threshold or credible-set filter was applied, and we pooled masses before taking the ratio instead of averaging dataset-specific ratios. The generating proportions were one for additive coding and zero for the other codings. Percentile 95% intervals used 500 bootstrap resamples of the 400 replication seeds, with shared resampling weights across conditions (bootstrap seed 20260911). The lead-PIP and coding-mass analyses summarize different quantities: a selected predictor's exact-match frequency and the distribution of total posterior mass, respectively. Neither analysis refitted SuSiE or tested the convergence or calibration of the weighted EM procedure.

Discussion

Implications and limits of the simulation evidence

These simulations show that an additive restriction can impair both causal-SNP recovery and credible-set coverage when causal effects are non-additive. The improvement is most pronounced for recessive effects, and remains apparent when comparison is based on SNP selection at similar empirical false discovery proportions. This supports genotype encoding as a consequential modeling choice, rather than treating a change in the number of credible sets as sufficient evidence of improved inference. At the same time, the expanded model incurs a modest credible-set recovery cost in additive settings and does not uniformly restore nominal coverage in weak, multicausal settings.

The coding-specific evaluation adds evidence that high-PIP dominant and recessive leads can identify the generating SNP and coding. High-PIP non-additive lead calls were infrequent under additive truth, providing a relevant control for the non-additive candidates in the GTEx analysis. This evidence concerns the unique global lead of an unweighted simulated fit; the GTEx summaries label individual credible sets by their lead predictor, so these numerical accuracy rates cannot be transferred directly to every reported set. Moreover, appreciable dominant and recessive PIP mass under additive truth means that coding-mass proportions are not estimates of biological effect prevalence without further validation. The present benchmark does not assess the weighted or iteratively reweighted fits.

The benchmark evaluates a defined family of alternatives: Gaussian phenotypes, a fixed sample size, at most five causal SNPs, and effects drawn from the same additive, dominant, and recessive encodings offered to SuSiE-mix. It therefore demonstrates the consequences of omitting these encodings and the benefit of including the correct family, rather than robustness to arbitrary nonlinear effects. Recessive effects were selected only when at least five minor-allele homozygotes were available; performance for rarer recessive effects is not established. Effect-size heterogeneity, interactions, untyped causal variants, and additional sample sizes also remain outside the present design.

Several distinctions matter for interpretation. Increased credible-set recovery can reflect larger sets, so comparative set sizes are needed before attributing every gain to improved resolution. Purity computed under different codings does not resolve this issue. The point-estimate comparisons of coverage, recovery, and lead accuracy also require uncertainty assessment that preserves shared seeds across methods and conditions. ROC and power-FDR curves assess discrimination; the additional lead-PIP reliability analysis evaluates exact SNP-coding matches and reveals remaining overconfidence, particularly for additive leads and some weak-signal settings. These descriptive checks do not establish formal false discovery control. Finally, the simulated outcome was not subjected to the GTEx expression transformations or covariate adjustment, and its fitting controls differed from the GTEx scan. The simulations make the real-data disagreements worth investigating, but do not by themselves validate an additional eQTL signal, a changed colocalization, or a particular regulatory mechanism.